TEXORA · LECTURE SHEET

In-depth Study of Rotor Spinning, Compact Spinning and Other Modern Short Staple Spinning Methods

Ring spinning twists and winds in one interrupted operation. The modern systems separate those jobs, individualise the fibres completely, or condense the spinning triangle away Rotor and compact spinning are studied in depth, with air-jet, vortex and friction spinning alongside.

YM-4201 · MODERN YARN MANUFACTURING · TOPIC 2 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. Why new spinning systems were developed

In ring spinning the spindle must rotate the whole package to insert twist and wind the yarn, so delivery stays around 20–30 m/min and winding is separate. Every modern system attacks one or more of those limits:

- **Rotor spinning** breaks the fibre stream into individual fibres, reassembles them in a rotor, and twists from the yarn end.
- **Compact spinning** keeps ring drafting and spindle but condenses the drafted strand before twist insertion, removing the spinning triangle.
- **Air-jet and vortex spinning** use swirling air to wrap surface fibres around a largely parallel core — not true twist throughout.
- **Friction spinning (DREF)** rolls a fibre fringe against suction drums to build a yarn, favouring coarse counts and technical yarns.

2. Rotor spinning in depth

2.1 Principle and machine sequence

Rotor spinning is an **open-end** system: the feed — a draw-frame sliver, not a roving — is opened back to individual fibres, so the fibre stream is genuinely broken. The fibres are transported by air into a small rotor spinning at very high speed. Centrifugal force throws them onto the rotor's inner wall; they slide into a V- or U-shaped collecting groove and form a fibre ring. A seed yarn, introduced through the

yarn withdrawal tube, touches the ring; rotor rotation twists the seed yarn back to the groove, the end continuously picks up new fibres, and the yarn is withdrawn and wound onto a large cross-wound package. Twisting and winding are fully separated, so packages can be large and no separate winding operation is needed.

1. **Feed.** Sliver is gripped by a feed roller against a spring-loaded feed plate.
2. **Opening.** A high-speed opening roller clothed with saw-tooth wire combs the sliver fringe into individual fibres. Trash, too heavy to follow the fibre air stream, is separated here.
3. **Transport.** Fibres travel in a transport channel, accelerated by suction air toward the rotor.
4. **Collection.** Fibres land on the sliding wall and are thrown into the collecting groove, stacking into a fibre ring.
5. **Formation and twist.** The rotor inserts one turn per revolution; the yarn end in the groove continuously incorporates the fibre ring.
6. **Withdrawal and winding.** Yarn passes through the navel and withdrawal tube to the separately driven package.

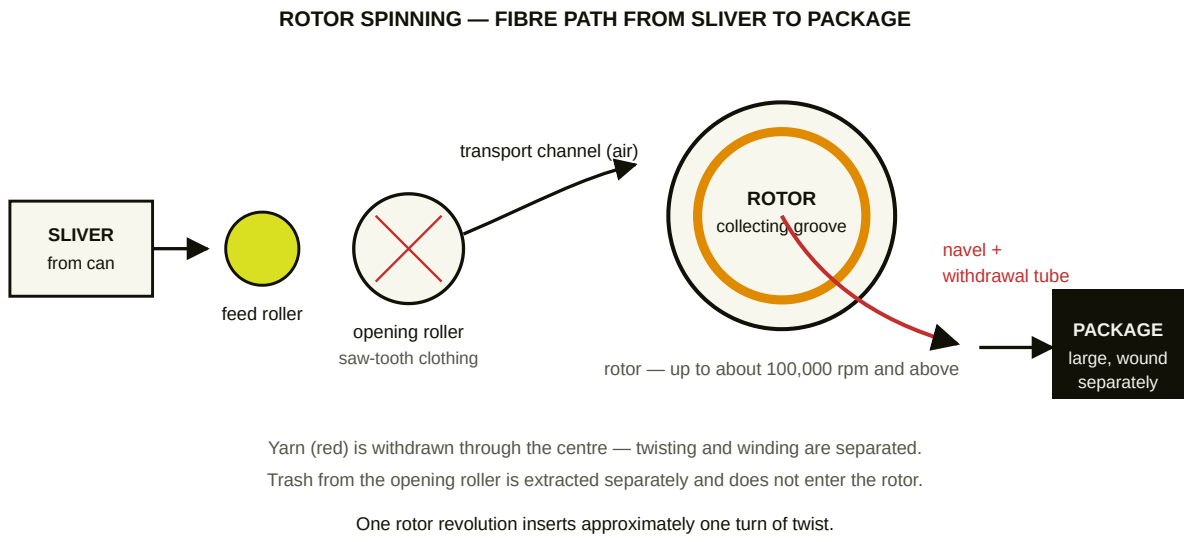


Figure 1 — Rotor spinning schematic. Sliver is opened to single fibres, transported by air into the rotor groove, twisted from the open end, and withdrawn through the navel to a separately wound package.

2.2 Rotor, groove, navel and false twist

The **rotor** is a small cup, a few tens of millimetres in diameter, on a precision bearing: smaller rotors run faster and suit finer counts, larger rotors suit coarser counts. The **collecting groove** shape governs how the fibre ring packs, and so yarn bulk. The **navel** is the funnel guide through which yarn leaves the rotor; yarn rubbing on it generates **false twist** — extra twist present between navel and collecting point while running but not fully retained in the delivered yarn. False twist strengthens the weak strand where

it peels from the groove and picks up fibres, reducing ends down; navel surface (smooth, grooved or structured ceramic) sets how much is produced.

2.3 Back-doubling and evenness

Fibres are laid onto the groove ring over many rotor revolutions before the corresponding yarn length is withdrawn. This cyclic layering of new fibres over fibres already in the groove is **back-doubling**. It averages short-term feed variation and explains rotor yarn's good short-term evenness despite sliver feed with no roving stage.

2.4 Rotor yarn structure and properties

A rotor yarn is not a miniature ring yarn. Its cross-section has three zones: a **core** of fibres laid with some twist but less well oriented than ring yarn (many arrive hooked or folded); a less ordered **sheath**; and characteristic **wrapper fibres** wound almost at right angles around the surface — a fingerprint of rotor spinning. Typical comparisons with ring yarn (textbook ranges):

- **Tenacity** lower, typically about 10–20% below ring yarn — fewer fibres are straight, parallel and fully load-sharing.
- **Evenness** good, often better than ring yarn in mass variation, with fewer sliver-origin imperfections.
- **Elongation and abrasion resistance** comparatively good; the yarn is bulky and uniform in diameter.
- **Hairiness** generally lower in long hairs, though wrapper fibres give a somewhat harsher surface.
- **Twist** must be set higher than in ring yarn for equivalent cohesion.

2.5 Limits and economics

Productivity per position is several times ring spinning's, labour per kilogram is lower, and roving and separate winding are eliminated. The limits are equally clear: as count fines, the number of fibres in the groove ring falls, stability drops and ends down rise, so rotor spinning is strongest in coarse to medium counts; and groove deposits of dust, finish and trash must be cleaned regularly. Denim, towels, blankets and coarse knits are natural rotor territory.

> **WORKED EXAMPLE — ROTOR DELIVERY AND PRODUCTION**
>
> Rotor speed = 100,000 rpm; yarn twist = 800 turns/m; yarn count = 30 tex.
>
> Each rotor revolution inserts one turn, so delivery = rotor speed / twist = 100,000 / 800 = **125 m/min**.
>
> Production per rotor per hour = delivery × 60 × tex / 1,000 = 125 × 60 × 30 / 1,000 = **225 g/h**.
>
> A ring spindle delivering 25 m/min of the same 30 tex yarn gives 25 × 60 × 30 / 1,000 = 45 g/h —

> the rotor position produces five times as much per hour in this example, before efficiency allowances.

3. Compact spinning in depth

3.1 The problem: the spinning triangle

At a ring frame the drafted ribbon leaves the front rollers over a finite width. Twist can only grip the fibres it reaches; edge fibres either escape twist as hairiness or are caught loosely and contribute little strength. The region between roller nip and twist consolidation point is the **spinning triangle** — the main source of ring hairiness. Compact spinning removes it by condensing the ribbon to a narrow, dense strand *before* twist reaches it, so essentially every fibre is incorporated under tension.

3.2 The condensing principle

After drafting the strand passes over a suction zone — a **perforated apron** or **lattice (mesh) apron** over a suction tube with an inclined slot, or a perforated drum. Negative pressure draws air through the strand; the angled flow compresses it sideways while pneumatic pressure holds fibres flat and close. The condensed strand, about as wide as the yarn itself, passes a delivery nip — also a twist stop — and twist consolidates it immediately, with no free triangle.

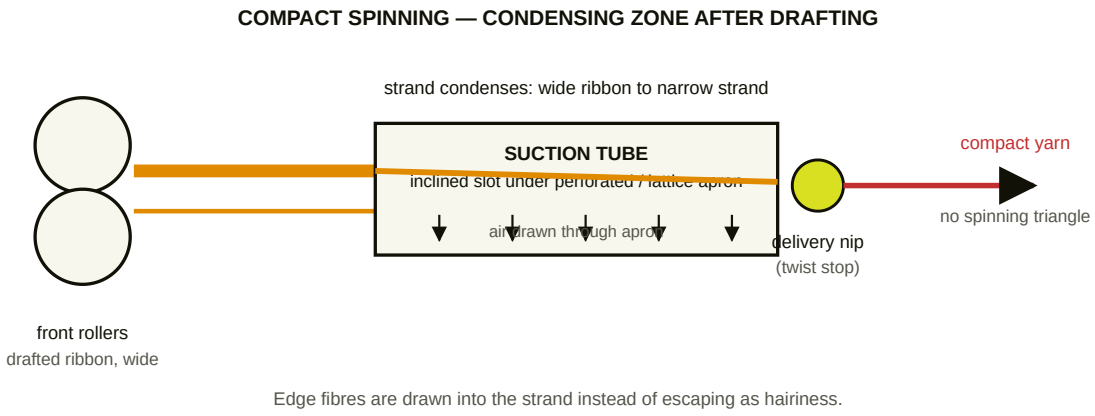


Figure 2 — Compact condensing zone. The drafted ribbon runs over a suction slot through a perforated or lattice apron, is condensed to yarn width, and is twisted immediately after the delivery nip.

3.3 Effects on yarn and fabric

- **Hairiness** falls sharply, especially the long hairs that cause weaving problems and a fuzzy fabric surface.
- **Tenacity** rises, typically around 5–15% over conventional ring yarn of the same material and count (typical textbook range), because more fibres are gripped by twist and sharing load.

- **Twist can be reduced** for the same strength target, giving a softer hand or higher productivity at a given strength — the mill chooses where to bank the gain.
- **Downstream gains:** fewer weaving ends down, clearer fabric surface, and in some applications less singeing and sizing.

The costs are the compact unit itself — suction energy, aprons as consumables, cleaning discipline — and unchanged delivery speed: compact spinning is a quality upgrade of ring spinning, not a productivity revolution. It is chosen for finer counts, premium shirting and high-quality knits.

4. Air-jet, vortex and friction spinning

In **air-jet spinning** a drafted strand passes through air nozzles whose jets create a swirling vortex. The principle is false twist plus wrapping: core fibres travel largely straight with little real twist, while edge fibres are wound around the core and hold the yarn together. The yarn is low in hairiness, somewhat stiff, with tenacity below ring yarn; it suits polyester-rich blends better than 100% cotton, whose short fibres wrap poorly.

Vortex spinning (MVS) refines the same family: the drafted bundle enters a chamber where a vortex of compressed air forms around a hollow spindle, and whirled fibre ends wrap the forming core as it is drawn through the spindle. The result is very low hairiness and good pilling behaviour at high delivery speed — again with a strength ceiling below good ring yarn.

In **friction spinning (DREF)**, opened fibres are fed onto the nip between two perforated suction drums whose surfaces roll the accumulating fibre fringe; friction supplies the twisting action and the yarn is withdrawn along the drum axis. The system is exceptionally tolerant: very coarse counts, short and recycled fibres, and core-sheath yarns. Yarns are bulky and weak for their count; the territory is coarse technical textiles, blankets, mops and carpet backing.

Glossary

Rotor groove — the collecting channel in the rotor wall where the fibre ring forms. **Navel** — the central yarn guide in a rotor unit whose friction generates false twist. **False twist** — twist present in a running yarn zone that does not remain in the delivered yarn. **Back-doubling** — cyclic layering of fibres in the rotor groove that averages short-term variation. **Wrapper fibres** — fibres wound nearly at right angles around a rotor yarn's surface. **Compact spinning** — ring spinning with pneumatic condensing before twist insertion. **Vortex spinning** — air-vortex wrapping of fibres around a parallel core.

Key points

• Rotor spinning separates twisting from winding: sliver in, large package out, no roving, no separate winding; delivery = rotor speed / twist.
• Back-doubling explains rotor yarn's good short-term evenness; hooked transport fibres and wrapper fibres explain its lower tenacity.
• Compact spinning condenses the strand by suction so the spinning triangle disappears — hairiness falls and tenacity rises, at ring-frame speeds.
• Friction (DREF) spinning tolerates coarse, short and recycled fibres and makes core-sheath yarns, at low yarn strength.

Practice

1. A rotor runs at 120,000 rpm and the yarn twist is 1,000 turns/m. Calculate the delivery speed, and the production per rotor per hour for a 25 tex yarn.
2. A rotor yarn is to be delivered at 140 m/min with a rotor speed of 98,000 rpm. What twist, in turns/m, will result?
3. Explain how false twist at the navel helps spinning stability although it does not remain in the delivered yarn.
4. A mill wants less hairiness and higher tenacity in a fine shirting yarn without changing its count range. Rotor or compact — choose one and justify the choice.
5. Describe the three zones of a rotor yarn and use them to explain its good evenness and its lower tenacity.

Answer key

1. Delivery = 120,000 / 1,000 = **120 m/min**. Production = 120 × 60 × 25 / 1,000 = **180 g/h per rotor**.
2. Twist = rotor speed / delivery = 98,000 / 140 = **700 turns/m**.
3. Navel friction adds temporary twist where the running yarn is weakest — just picking up fibres — so it resists breaking there; leaving the zone, the false component untwists and delivered twist is unchanged.
4. Compact spinning. Fine shirting counts are beyond rotor spinning's practical fine limit, and rotor yarn's lower tenacity and wrapper-fibre surface would harm a premium fabric. Compact keeps the ring count range while reducing hairiness and raising tenacity.
5. Core: fibres with twist but poorer orientation (many arrive hooked or folded). Sheath: less ordered still. Wrappers bind the surface at right angles and share little load. Evenness is good because back-doubling averages feed variation; tenacity is lower because fewer fibres are straight and fully load-sharing.